Generators of Strongly Continuous Semigroups

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1. Introduction

In the forties of the 20^{th} century the cornerstones of semigroup theory has been developed independently by the mathematicians E. HILLE and K. YOSIDA. Both proved in different ways a theorem that clarifies the condition, when a linear operator A acting in a Banach space X is the infinitesimal generator of a strongly continuous semigroup $\{S(t)\}_{t\geq 0}$ of bounded linear operators.

This subject is of great importance with respect to the existence and uniqueness of solutions for the abstract initial-value problem

$$\dot{x} + Ax = 0, \quad x(0) = x_0 \in X,$$

which is meaningful for evolutional differential equations posed in a domain $G \subset \mathbb{R}^d$ and governed by linear-elliptic partial differential operators.

2. Strongly Continuous Semigroups

Let X be a complex Banach space and $\mathscr{L}(X)$ the unital Banach algebra of bounded linear operators $S : X \to X$ with identity 1_X. Then by a *C_0-semigroup* or a *strongly continuous semigroup* of bounded linear operators $\{S(t)\}_{t\geq 0}$ is meant an operator-valued mapping

$$S : [0, \infty) \to \mathscr{L}(X),$$

shortly denoted $\{S(t)\}_{t\geq 0}$, which distinguishes by the conditions

(G-1) $S(t+s) = S(t) \circ S(s)$ for all $t, s \geq 0$ (the *semigroup property*);
(G-2) $\lim_{t\downarrow 0} S(t)\, x = x$ for all $x \in X$.

It follows from (G-2), which is nothing but strong continuity from the right in $t = 0$, that $S(0) = 1_X$. The conditions (G-1) and (G-2) together imply continuity from the right of the semigroup even in each $t > 0$. In case of $X = \mathbb{R}$, both properties are shared by the

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family $\{\mathcal{E}(t)\}_{t\geq 0}$ of exponential functions $\mathcal{E}(t):\lambda\in\mathbb{R}\mapsto e^{-\lambda t}$, hence $\{\mathcal{E}(t)\}_{t\geq 0}$ serves as an elementary example of a strongly continuous semigroup.

For a given Banach space X the totality of strongly continuous semigroups may be classified as follows: for each such $\{S(t)\}_{t\geq 0}$ there are constants $M\geq 0$ and $\omega\in\mathbb{R}$ such that

$$\|S(t)\| \leq Me^{\omega t} \quad \text{for } t>0. \tag{2.1}$$

In fact, let $M\geq 1$ such that $\|S(s)\|\leq M$ holds for all $s\in[0,1]$. For $t>0$ there is $s\in[0,1)$ and $n\in N$ such that $t=n+s$. Then

$$\|S(n)\| \leq M\,\|S(n-1)\|$$

for $n>1$ and

$$\|S(t)\| \leq \|S(n)\|\cdot\|S(s)\| \leq M^{n+1} = Me^{n\cdot\ln M}.$$

In the special case $(M,\omega)=(0,1)$ and $\|S(t)\| \leq 1$ for $t\geq 0$, respectively, the semigroup is said to be *contracting.*

By the *orbit map* or the *trajectory* η_x of a C_0-semigroup $\{S(t)\}_{t\geq 0}$ associated with an initial element $x\in X$ we mean the vector-valued mapping $\eta_x:[0,\infty)\to X$ defined by $\eta_x(t):=S(t)\,x$. Both conditions (G-1) and (G-2) together imply that η_x is continuous. But the orbit map is even differentiable under a very mild assumption. In fact, η_x is differentiable in $(0,\infty)$ for each $x\in X$ iff it is differentiable from the right in $t=0$ only.

3. Infinitesimal Generators

If $\{S(t)\}_{t\geq 0}$ is a strongly continuous semigroup, then its *generating operator* or *infinitesimal generator* (briefly denoted the *generator*) is defined as a mapping A in X by means of the graph

$$\Gamma_A := \{(x,y)\in X\times X : y = \lim_{h\downarrow 0}\frac{1}{h}(S(h)x-x) \text{ exists}\}.$$

Hence the *domain of definition* $\mathfrak{D}(A)$ of the mapping A consists of those elements $x\in X$, for which the orbit map η_x is differentiable from the right in $t=0$, or equivalently,

$$\mathfrak{D}(A) := \{x\in X : \eta_x:[0,\infty)\to X \text{ is differentiable}\}.$$

It is not hard to show that $\mathfrak{D}(A)$ is a linear subspace of X and the mapping A is linear, so A constitutes a "linear operator" in X.

The following theorem contains a couple of assertions, which will later be used to prove the main theorems of this text.

THEOREM 3.1. Let $\{S(t)\}_{t\geq 0}$ be a C_0-semigroup with infinitesimal generator A. Then

(i) For $x \in X$

$$x_t := \lim_{h\to 0} \frac{1}{h}\int_t^{t+h} S(s)x\, ds = S(t)x. \tag{3.1}$$

(ii) For $x \in X$ and $\int_0^t S(s)x\, ds \in \mathfrak{D}(A)$

$$A\int_0^t S(s)x\, ds = S(t) - x. \tag{3.2}$$

(iii) For $x \in \mathfrak{D}(A)$ and $S(t)x \in \mathfrak{D}(A)$

$$\frac{d}{dt}S(t)x = A\,S(t)x = S(t)\,Ax. \tag{3.3}$$

(iv) For $x \in \mathfrak{D}(A)$

$$S(t)x - S(s)x = \int_s^t S(\tau)\,Ax\, d\tau = \int_s^t A\,S(\tau)x\, d\tau. \tag{3.4}$$

PROOF: Assertion (i) follows from the continuity of the orbit mapping η_x. To show assertion (ii), let $x \in X$ and $h > 0$. We have

$$\frac{S(h) - 1_X}{h}\int_0^t S(s)x\, ds = \frac{1}{h}\int_0^t S((s+h) - S(s)x)\, ds =$$
$$\frac{1}{h}\int_t^{t+h} S(s)x\, ds - \frac{1}{h}\int_0^h S(s)x\, ds.$$

Now, taking $h \downarrow 0$, the right hand side tends to $S(x) - x$ whence (ii) follows.

To prove (iii), let $x \in \mathfrak{D}(A)$ and $h > 0$. Then

$$\frac{S(h) - 1_X}{h}S(t)x = S(t)\frac{S(h) - 1_X}{h}x \to S(t)\,Ax \text{ as } h \downarrow 0. \tag{3.5}$$

Hence $S(t)x \in \mathfrak{D}(A)$ and $A\,S(t)x = S(t)Ax$. From (3.5) we also deduce that

$$\frac{d^+}{dt}S(t)x = AS(t)x = S(t)\,Ax.$$

Now we need to verify that for $t > 0$ the left derivative exists and equals to $S(t)\,Ax$. To do so, we write

$$\lim_{\downarrow 0} \Big(\frac{S(t)x - S(t-h)x}{h} - S(t)\,Ax\Big) =$$

$$\lim_{\downarrow 0} S(t-h)\Big(\frac{S(h)x - x}{h} - Ax\Big) + \lim_{\downarrow 0} \big(S(t-h)\,Ax - S(t)\,Ax\big)$$

and notice that both terms of the last line are zero. The first one due to the fact that $x \in \mathfrak{D}(A)$ and $\|S(t-h)\|$ is bounded on $0 \le h \le t$, and the second one because of property (G-2) of $S(t)$, which implies assertion (iii).
Finally, assertion (iv) is deduced by integrating (3.3) from 0 to t. □

Let us turn to the concept of densely defined and closed linear operator. First, a linear operator A is said to be *densely defined* in X, if its domain of definition $\mathfrak{D}(A)$ lies dense in X. Second, the operator A is called *closed*, if the set

$$\Gamma_A := \{\,(x, Ax) : x \in \mathfrak{D}(A)\,\} \subset X \times Y,$$

denoted the *graph* of A, forms a closed subset of the space $X \times Y$ with respect to the product topology $\mathcal{T}_{X\times Y}$. This property especially implies that the *kernel* $\mathfrak{N}(A) := \{x \in \mathfrak{D}(A) : Ax = 0\}$ is closed.

It is a matter of fact that a linear operator A acting between normed linear spaces is closed iff it is *sequentially closed*, that is, $x_n \to x$ and $Ax_n \to y$ implies $x \in \mathfrak{D}(A)$ and $Ax = y$, briefly written

$$\left.\begin{array}{r} x_n \to x \\ Ax_n \to y \end{array}\right\} \implies x \in \mathfrak{D}(A) \text{ and } Ax = y.$$

This can be verified by the fact that a subset of a normed linear space is closed if and only if it is sequentially closed. Checking the sequentially closeness is the usual method to prove that a given linear operator is graph-closed.

We finally mention that if A is closed, then for $\zeta \in \mathbb{C}$ the operator $\zeta - A$ with domain of definition $\mathfrak{D}(\zeta - A) = \mathfrak{D}(A)$ is closed, too. Moreover, if A is closed and bijective, then the inverse operator $A^{-1} : \mathfrak{R}(A) \subset X \to \mathfrak{D}(A) \subset X$ is also linear and closed. The assertion of A^{-1} being closed is based on the facts that
(1) if Γ_A is a closed set in $X \times Y$, then $\Gamma_{A^{-1}}$ is closed in $Y \times X$ and
(2) the permutation $(x, y) \mapsto (y, x)$ is a homeomorphism between the spaces $X \times Y$ and $Y \times X$.

Our next theorem indicates that infinitesimal generators fit into the class of densely defined and closed linear operators:

THEOREM 3.2. If A is the generator of a strongly continuous semigroup, then A is densely defined and closed.

PROOF: To show that A is densely defined, consider the value x_t as defined in (3.1). From assertion 3.1 (ii) we deduce $x_t \in \mathfrak{D}(A)$ for all $t > 0$ and by 3.1 (i) we have $x_t \to x$ as $t \downarrow 0$, which implies that the closure of $\mathfrak{D}(A)$ is the whole of X.

To show that A is closed, let $x_n \in \mathfrak{D}(A)$ with $x_n \to x$ and $Ax_n \to y$. By 3.1 (iv) we have

$$S(t)x_n - x_n = \int_0^t S(t)\, Ax_n \, ds, \tag{3.6}$$

where the integrand converges to $S(s)y$ uniformly on bounded intervals. Thus letting $n \to \infty$ in (3.6), we obtain

$$S(t)x - x = \int_0^t S(t)y \, ds.$$

Dividing the last identity by $t > 0$ and making $t \downarrow 0$, we deduce by employing 3.1 (i) that $x \in \mathfrak{D}(A)$ and $Ax = y$. □

Using the fact that infinitesimal generators are densely defined, we may quote the following uniqueness theorem:

THEOREM 3.3. If A and B are generators of C_0-semigroups, then $A = B$ imply that the semigroups generated by them are identical.

PROOF: Let $\{T(t)\}_{t\geq 0}$ and $\{S(t)\}_{t\geq 0}$ be the generated semigroups of A and B, respectively, and let $x \in \mathfrak{D}(A) = \mathfrak{D}(B)$.
By 3.1 (iii), the function $s \mapsto T(t-s)\, S(s)x$ is differentiable and

$$\begin{aligned} \frac{d}{ds} T(t-s)\, S(s)x &= -A\, T(t-s)\, S(s)x + T(t-s)\, B\, S(s)x \\ &= -T(t-s)\, A\, S(s)x + T(t-s)\, B\, S(s)x = 0 \end{aligned}$$

holds. Therefore the function $s \mapsto T(t-s)\, S(s)x$ is constant and in particular its values at $s = 0$ and $s = t$ are the same, that is, $T(t)x = S(t)x$. This is true for any $x \in \mathfrak{D}(A)$, since by theorem 3.2 the domain of definition $\mathfrak{D}(A)$ is dense in X and $T(t)$ and $S(t)$ are bounded, thus $T(t)x = S(t)x$ for every $x \in X$. □

4. C_0-Semigroups and Cauchy Problems

The *homogeneous Cauchy problem* of first order associated with a densely defined and closed linear operator A acting in a Banach space X is to find a solution of the initial value problem

$$\begin{cases} \dot{x}(t) &= A\,x(t), \\ x(0) &= x_0 \in X. \end{cases} \tag{4.1}$$

The task is to find *vector-valued* solutions $x:[0,\tau)\to X$ of $\dot{x}=Ax$ with $x(0)=x_0$.

Remember that a vector-valued function $x:[a,b]\to X$ is called *classical differentiable* in $t_0\in(a,b)$, if there is an element $x'(t_0)\in X$ such that

$$\lim_{t\to t_0}\left\|\frac{x(t)-x(t_0)}{t-t_0}-x'(t_0)\right\| = 0.$$

Moreover, the function x is called classical differentiable in (a,b), if x is so in each $t\in(a,b)$. The linear space consisting of functions having *continuous* classical derivatives in (a,b) is denoted $C^1(a,b;X)$.

The spaces $C^k(a,b;X)$ $(k\in\mathbb{N})$ are defined recursively, and the space $C^\infty(a,b;X)$ is defined by taking the intersection of all spaces $C^k(a,b;X)$, $k\in\mathbb{N}_0$.

Concepts of solution for problem (4.1) are given as follows:

1. A vector-valued function $x\in C([0,\tau),\mathfrak{D}(A))\cap C^1((0,\tau),X)$ is called a *classical solution* of (4.1), if it satisfies the initial condition $x(0)=x_0$ and the differential equation in each point $t\in(0,\tau)$;
2. a continuous function $x:[0,\tau)\to X$ is said to be a *mild solution* of (4.1) if

$$(1)\ \ \tilde{x}(t) := \int_0^t x(s)\,ds \in\mathfrak{D}(A) \quad\text{and}\quad (2)\ \ A\,\tilde{x}(t)=x(t)-x_0$$

holds for all $0<t<\tau$, where the integral on the right-hand side of (1) is meant in the Riemannian sense. One is led to the notion of mild solution in a natural way by integrating both sides of the equation $\dot{x}=A\,x$, which then becomes into

$$\int_0^t \dot{x}ds = x(t)-x(0) = \int_0^t A\,x(s)\,ds.$$

If A is closed, then the action of A and performing the integration commutes for continuous functions $x:[0,t]\to\mathfrak{D}(A)$, that is,

$$\int_0^t A\,x(s)\,ds \;=\; A\int_0^t x(s)\,ds,$$

and to the end we obtain the equation

$$x(t) \;=\; x(0) + A\int_0^t x(s)\,ds \qquad (t \in (0,\tau)).$$

Notice that each classical solution is a mild solution, and reversely a mild solution is classical iff it is continuously differentiable in $(0,\tau)$. Thus classical and mild solutions of homogeneous Cauchy problems only differ with respect to this regularity property.

After having the notions of classical and mild solution for the homogeneous Cauchy problem available, we want to introduce the concepts of *correct posedness* and *well-posedness*.

First, the homogeneous Cauchy problem (4.1) is called *correctly posed*, if there is a unique classical solution x for any initial value $x_0 \in \mathfrak{D}(A)$. Second, it is denoted *well-posed*, if it is correctly posed and the solution x depends continuously on the initial value x_0, or equivalently, if for each $\tau > 0$ there is $c_\tau > 0$ such that

$$\|x(t)\| \le c_\tau\, \|x_0\|$$

for all $t \in (0,\tau)$ and all $x_0 \in X$.

In [1] the continuous dependency of the solutions from the initial values is expressed as follows: if $\{x_n\}_{n\in\mathbb{N}}$ is a converging sequence in $\mathfrak{D}(A)$ with limit element $0 \in X$, then the related solutions $t \mapsto x_n(t)$ are uniformly convergent to the zero function in all compact intervals $[0, t_0]$ for $0 < t_0 < \tau$.

We mention without proof that problem (4.1) is well-posed iff it is correctly posed and the operator A is *regular*, that is, $\rho(A) \ne \emptyset$ holds (for the definition of the set $\rho(A)$ see section 5).

The main theorem of this section provides the following equivalence and emphasizes the meaning of C_0-semigroups for the solution theory of abstract linear evolution equations:

THEOREM 4.1. The Cauchy problem (4.1) is well-posed iff the linear operator A is the infinitesimal generator of a C_0-semigroup.

For the proof of this theorem see [1], theorem II.6.6 at p. 112.

5. Resolvent Functions

In this section we want to introduce some required concepts of spectral theory. Any complex number $\zeta \in \mathbb{C}$ is called a *regular value* of the operator A, if $A - \zeta : \mathfrak{D}(A) \to X$ is bijective and the inverse operator

$$R(A, \zeta) := (\zeta - A)^{-1} : X \to \mathfrak{D}(A)$$

is continuous. The set $\rho(A)$ of regular values is denoted the *resolvent set* of A. If $\zeta \in \rho(A)$, then $R(A, \zeta)$ is everywhere defined in X. Since we assume A to be closed, the operators $\zeta - A$ and $R(A, \zeta)$ are closed, too, and the *closed graph theorem* implies that $R(A, \zeta)$ is continuous, that is, $R(A, \zeta) \in \mathscr{L}(X)$[1]. The linear operator $R(A, \zeta)$ is called the *resolvent* of A with respect to $\zeta \in \rho(A)$, and the operator-valued mapping

$$\begin{cases} R(A, \cdot) : \rho(A) \to \mathscr{L}(X)\,, \\ \zeta \mapsto (\zeta - A)^{-1} \end{cases} \tag{5.1}$$

is denoted the *resolvent function* (or shortly the *resolvent*) of A.

But the resolvents of a linear operator A are not independent from each other. In particular, for all $\zeta_1, \zeta_2 \in \rho(A)$ the *first resolvent equation*

$$R(A, \zeta_1) - R(A, \zeta_2) = (\zeta_2 - \zeta_1)\, R(A, \zeta_1) \circ R(A, \zeta_2) \tag{5.2}$$

holds, which implies that the resolvents $R(A, \zeta_1)$ and $R(A, \zeta_2)$ commute with each other.

Let us turn to the resolvent function of a linear operator A with non-empty resolvent set $\rho(A)$. In fact, if $\zeta_0 \in \rho(A)$, we have $\zeta \in \rho(A)$ for all $\zeta \in \mathbb{C}$ satisfying

$$|\zeta - \zeta_0| < r := \|R(A, \zeta_0)\|^{-1}\,,$$

thus $\rho(A)$ is an open set in $\mathbb{C}$. For values ζ in the disk with center ζ_0 and radius r one may represent the resolvent function $R(A, \cdot)$ as a power series with coefficients in the operator algebra $\mathscr{L}(X)$:

$$R(A, \zeta) = \sum_{n=0}^{\infty} R(A, \zeta_0)^{n+1}\, (\zeta_0 - \zeta)^n.$$

This implies that $R(A, \cdot)$ is locally holomorphic in $\rho(A)$.

[1] If A is *not* closed, one has in addition to require that for $\zeta \in \rho(A)$ the inverse operator $R(A, \zeta)$ is bounded.

Our next theorem ephasizes the meaning of operators to be assumed as closed:

THEOREM 5.1. If $\rho(A) \neq \emptyset$, then A is closed.

PROOF: Let $\zeta \in \rho(A)$ such that $A - \zeta$ is bijective with domain of definition $\mathfrak{D}((A-\zeta)^{-1}) = Y$ and range $\mathfrak{R}((A-\zeta)^{-1}) = \mathfrak{D}(A)$. Let $x_n \to x$ with $x_n \in \mathfrak{D}(A)$ for $n \in \mathbb{N}$ and $Ax_n \to y$. Then

$$x = \lim_{n\to\infty} x_n = \lim_{n\to\infty} (\zeta - A)^{-1}(\zeta - A)x_n = (\zeta - A)^{-1}(\zeta x - y). \quad (5.3)$$

Thus $x \in \mathfrak{R}((\zeta - A)^{-1}) = \mathfrak{D}(A)$. When applying $(\zeta - A)^{-1}$ from left to (5.3), we obtain $\zeta x - Ax = \zeta x - y$, hence $Ax = y$, which is nothing but closeness of A. □

Thus a reasonable spectral theory makes only sense for linear operators being at least continuous or closed. Those operators, which are closed and in addition own non-empty resolvent sets, are often called *regular*.

In the second part of this section we want to establish a relationship between strongly continuous semigroups and the resolvents of their infinitesimal operators, denoted the *Laplace transform* of the semigroup.

Let the semigroup $\{S(t)\}_{t\geq 0}$ be strongly continuous. Then for fixed $x \in X$ and $\zeta \in \mathbb{C}$, the value $L_n(\zeta, x)$ defined by

$$L_n(\zeta, x) := \frac{1}{(n-1)!} \int_0^\infty t^{n-1} e^{-\zeta t} S(t)x \, dt \quad (5.4)$$

is called the (right side) *Laplace transform* of order n of the semigroup with respect to x and ζ. The expressions have to be understood as improper vector-valued Riemann integrals. But we cannot expect that $L_n(\zeta, x)$ exists without further assumptions.

The following theorem describes necessary conditions for a linear operator A to be the infinitesimal generator of a C_0-semigroup:

THEOREM 5.2. Let $\{S(t)\}_{t\geq 0}$ be a strongly continuous semigroup of type (M, ω) and with infinitesimal generator A. Then:

(a) If $\operatorname{Re}\zeta > \omega$, then $\zeta \in \rho(A)$ and $R(A,\zeta)^n x = L_n(\zeta, x)$ for all $n \in \mathbb{N}$ and $x \in X$.

(b) For all $n \in \mathbb{N}$ and $\operatorname{Re}\zeta > \omega$ the following resolvent estimates hold:

$$\|R(A,\zeta)^n\| \le M(\operatorname{Re}\zeta-\omega)^{-n}.$$

PROOF: The values $L_n(\zeta,x)$ are well-defined for $\operatorname{Re}\zeta > \omega$ and $x\in X$. Hence the mapping $L_n(\zeta)x := L_n(\zeta,y)$ forms a linear operator in X with norm estimate

$$\|L_n(\zeta,\cdot)\| \le \frac{1}{(n-1)!}\int_0^\infty t^{n-1}e^{-\operatorname{Re}\zeta t}e^{\omega t}\,dt = \frac{M}{(\operatorname{Re}\zeta-\omega)^n}.$$

holds. For $n=1$ we have $L_1(\zeta)Ax = -x+\zeta L_1(\zeta)x$, whereas for $n>1$ the equality $L_n(\zeta)Ax = -L_{n-1}(\zeta)x+\zeta L_1(\zeta)x$ holds. Applying $(S(h)-1_X)/h$ to equation (5.4), we obtain by taking the limit $h\to 0$:

$$\begin{cases} A\,L_1(\zeta)x = -x+\zeta L_1(\zeta)x \\ A\,L_n(\zeta)x = -L_{n-1}(\zeta)x+\zeta L_n(\zeta)x \quad (n>1), \end{cases}$$

which leads to

$$\begin{cases} L_1(\zeta)(\zeta-A) = (\zeta-A)L_1(\zeta) = 1_X, \\ L_n(\zeta)(\zeta-A) = (\zeta-A)L_n(\zeta) = L_{n-1}(\zeta). \end{cases}$$

In conclusion, the resolvent $R(A,\zeta)$ exists and for all $n\in\mathbb{N}$ we have $R(A,\zeta)^n = L_n(\zeta)$. Especially for $\operatorname{Re}\zeta > \omega$ we have $(\omega,\infty)\subset\rho(A)$. □

Let us put together the necessary conditions for the linear operator A to be an infinitesimal generator:

COROLLARY 5.1. In order A to be the generator of a C_0-semigroup of type (M,ω), the following properties must be fulfilled.
(a) A is densely defined and closed.
(b) $(\omega,\infty)\subset\rho(A)$ and

$$\|R(A,\zeta)^n\| \le M(\zeta-\omega)^{-n} \quad \text{for } \zeta\in(\omega,\infty) \text{ and } n\in\mathbb{N}. \tag{5.5}$$

The family of inequalities quoted in (b) of this corollary is often denoted the *Hille-Yosida condition.*

In the case, where $(M,\omega)=(1,\omega)$, that is, the semigroup is *quasi-contractive*, we have $\|R(A,\zeta)\| \le (\zeta-\omega)^{-1}$ for $\zeta>\omega$ $(n=1)$, which implies that all the conditions for $n>1$ are also fulfilled. Hence the family of conditions in (5.5) reduces to a single one, and this can often verified very easily. Due to the importance of this fact, we quote this special case in a further corollary:

COROLLARY 5.2. In order A to be the generator of a C_0-semigroup of type $(1,\omega)$, the following properties must be fulfilled.

(a) A is densely defined and closed.
(b) $(\omega,\infty) \subset \rho(A)$ and

$$\|R(A,\zeta)\| \le \frac{1}{\zeta-\omega} \quad \text{for } \zeta \in (\omega,\infty). \tag{5.6}$$

6. Generator Theorems of Hille-Yosida Type

As it has already been elaborated in the third section, to each strongly continuous semigroup $\{S(t)\}_{t\ge 0}$ of bounded operators there is a linear operator A as its infinitesimal generator. Reversely, the question arises, under which conditions a linear operator constitutes the infinitesimal generator of a C_0-semigroup. For bounded operators the answer can be given quickly: if $\|A\|$ is the operator norm of A, then the mappings

$$\exp(A)(t)\,x := \sum_{n=0}^{\infty} \frac{(tA)^n}{n!}\,x$$

for $x \in X$ and $t > 0$ form a strongly continuous semigroup.

The general answer, however, is given by the *Feller-Miyadera-Hille-Yosida theorem*:

THEOREM 6.1. Let A be a linear operator acting in a complex Banach space X with domain of definition $\mathfrak{D}(A)$. Then the following assertions are equivalent to each other:

(a) The operator A generates a C_0-semigroup of type (M,ω).
(b) the operator A is densely defined and closed, the resolvent set of A fulfills $(\omega,\infty) \subset \rho(A)$ and for $\zeta > \omega$ the resolvent $R(A,\zeta)$ satisfies the Hille-Yosida conditions

$$\|R(A,\zeta)^n\| \le M(\zeta-\omega)^{-n} \quad (n \in \mathbb{N}). \tag{6.1}$$

PROOF: It was already shown in the last section that (a) implies (b) (see corollary 5.1). Reversely, it is to prove that (b) implies (a). First of all, we shall reformulate and prove theorem 6.1 for the case of quasi-contractive semigroups and then come back to the proof of the implication (b) $\Rightarrow$ (a) at the end of this section.

THEOREM 6.2. A linear operator A is the infinitesimal generator of a C_0-semigroup $\{S(t)\}_{t\geq 0}$ of type $(1,\omega)$ iff

(a) A is densely defined and closed.

(b) (ii) The resolvent set $\rho(A)$ contains the interval $I := (\omega, \infty)$, and for $\zeta \in I$ the estimate $\|R(A,\zeta)\| \leq (\zeta - \omega)^{-1}$ holds.

PROOF: If A is an infinitesimal generator, then the validness of conditions (a) and (b) was already quoted in corollary 5.2.

Reversely, we want to sketch how the assertions (a) and (b) imply that A generates a quasi-contractional C_0-semigroup. First of all, it can be verified that (5.5) implies $\lim_{\zeta\to\infty} \zeta R(A,\zeta) = x$ for all $x \in \mathfrak{D}(A)$. As a consequence, the so-called *Yosida operators*

$$A_n := nA(n-A)^{-1} = n^2(A-n)^{-1} - n$$

approximate the operator A as $n \to \infty$ in the sense that

$$\lim_{n\to\infty} A_n x = Ax$$

for all $x \in \mathfrak{D}(A)$. Since the operators A_n are bounded, one may define semigroups for each of them by letting

$$e^{tA_n}\, x := \sum_{k=0}^{\infty} \frac{(tA_n)^k}{k!}\, x \qquad (x \in X).$$

It follows that $\|e^{tA_n}\| \leq 1$ for $t > 0$, so we obtain even semigroups of contractions. Now the main part of the proof is to show that this sequence of semigroups is Cauchy and the semigroup generated by the operator A is given by

$$S(t)x := \lim_{n\to\infty} e^{tA_n}\, x \qquad (x \in X). \tag{6.2}$$

To be precise, one first has to verify property (S-1) and (S-2) and second the operator A to be the infinitesimal generator of the semigroup (6.2). □

We now come back to the proof of theorem 6.1 by using the assertion of the already proved theorem 6.2. □

Recall that in the more special case, where $M = 1$ and $\omega = 0$, the semigroup of type $(1,0)$ is called a *semigroup of contractions.* From theorem 6.2 we deduce the following

COROLLARY 6.1. Let A be a linear operator acting in a complex Banach space X with domain of definition $\mathfrak{D}(A)$. Then the following assertions are equivalent to each other:

(a) A generates a strongly continuous semigroup of type $(1,0)$.
(b) A is densely defined and closed, the resolvent set of A fulfills $(0,\infty)\subset\rho(A)$ and the resolvent function $R(A,\zeta)$ satisfies

$$\|R(A,\zeta)\| \le \zeta^{-1} \quad (\zeta>0).$$

The statement $(0,\infty)\subset\rho(A)$ in this corollary means that one has to verify the existence and uniqueness of a solution for each stationary equation $Ax-\zeta x=y$, where $\zeta>0$ and $y\in X)$ is arbitrary.

7. The Lumer-Phillips Theorem

Preliminary, the first theorem in this section deals with operators having the *bounded-below* property.

A linear operator A acting in a Banach space X is called *bounded below* if there is $\theta>0$ such that

$$\|Ax\|_Y \ge \theta\,\|x\|_X \text{ for all } x\in\mathfrak{D}(A). \tag{7.1}$$

The *bounded-below theorem* is a fundamental assertion in operator theory and is helpful to verify the closeness of $\mathfrak{R}(A)$:

THEOREM 7.1. If A fulfills (7.1), then A is injective and the operator $A^{-1}:\mathfrak{R}(A)\subset X\to X$ is bounded with operator norm $\|A^{-1}\|\le\theta^{-1}$. Furthermore, the range $\mathfrak{R}(A)$ is closed if and only if A is closed.

PROOF: Injectivity of the operator A directly follows from (7.1). To show that $\mathfrak{R}(A)$ is closed, let $y_n\to y$ and $y_n\in\mathfrak{R}(A)$ for $n\in\mathbb{N}$. Then for each $n\in\mathbb{N}$ there is $x_n\in\mathfrak{D}(A)$ such that $y_n=Ax_n$. In fact, $\{x_n\}$ is a Cauchy sequence by (7.1) with limit element $x\in X$. Since A is closed, we have $x\in\mathfrak{D}(A)$ and $Ax=y$. Hence $y\in\mathfrak{R}(A)$ and $\mathfrak{R}(A)$ is closed.
The inverse operator $A^{-1}:\mathfrak{R}(A)\to X$ is closed as well, hence by the closed graph theorem it is bounded with operator norm θ^{-1}. □

For the rest of this section we switch to the Hilbert space setting and assume $X=H$ to be a complex Hilbert space.

A linear operator A acting in H is said to be *dissipative*, if the numerical range $\Theta(A):=\{\langle Ax,x\rangle : x\in\mathfrak{D}(A),\|x\|=1\}$ fulfills

$$\Theta(A)\subseteq\mathbb{C}^- := \{\zeta\in\mathbb{C}:\operatorname{Re}\zeta\le 0\},$$

or equivalently, if $\operatorname{Re}\langle Ax,x\rangle\le 0$ for all $x\in\mathfrak{D}(A)$. Furthermore, the operator A is said to be *quasi-dissipative*, if there is $\lambda_0\in\mathbb{R}$ such that $A+\lambda_0$ is dissipative.

The next theorem describes dissipativity by another condition:

THEOREM 7.2. $\operatorname{Re}\langle Ax, x\rangle \leq 0$ for $x \in \mathfrak{D}(A)$ is equivalent to

$$\|(\zeta - A)x\| \geq \zeta\,\|x\| \qquad (\zeta > 0). \tag{7.2}$$

PROOF: For $x \in \mathfrak{D}(A)$ the inequality

$$\|\zeta x - Ax\|^2 - \|\zeta x\|^2 \geq -2\zeta \operatorname{Re}\langle Ax, x\rangle \geq 0$$

holds, which implies (7.2). □

Property (7.2) implies that the operator $\zeta - A$ is bounded below, hence $\zeta - A$ is injective and $\mathfrak{R}(\zeta - A)$ is closed.

A dissipative operator A is said to be *m-dissipative*, if it does not have any proper dissipative extension. Equivalently, a dissipative operator A is m-dissipative, if there is $\zeta_0 > 0$ such that $\zeta_0 - A$ is surjective. The property $\mathfrak{R}(\zeta_0 - A) = X$ is called the *range condition* for a m-disspative operator.

The following theorem is used in the proof of theorem 7.5 below:

THEOREM 7.3. If A is a m-dissipative linear operator, then A is densely defined in H and closed.

PROOF: To verify that A is densely defined, let $y \in \mathfrak{D}(A)^{\perp}$ and $x \in \mathfrak{D}(A)$ be a vector such that $x - Ax = y$. Surely, such a vector exists since $1 - A$ is surjective. Then we compute

$$0 = \langle x, y\rangle = \langle x, x - Ax\rangle = \langle Ax, x\rangle - \|x\|^2 \leq 0,$$

which implies $x = 0$ and $y = x - Ax = 0$. Hence $\mathfrak{D}(A)^{\perp} = \{0\}$ and consequently $\overline{\mathfrak{D}(A)} = H$.

Two show that A is closed, we have to verify that if $x_n \to x$ and $Ax_n \to y$, then $x \in \mathfrak{D}(A)$ and $Ax = y$. Clearly, $(\zeta_0 - A)x_n \to \zeta_0 x - y$, hence

$$x = \lim_{n\to\infty} x_n = \lim_{n\to\infty} (\zeta_0 - A)^{-1}(\zeta_0 - A)\, x_n = (\zeta_0 - A)^{-1}(x - y)$$

so $x = (\zeta_0 - A)^{-1}(x - y) \iff \zeta_0 x - Ax = \zeta_0 x - y \iff Ax = y$. □

Our next theorem provides a very useful property for dissipative operators:

THEOREM 7.4. If A is m-dissipative, then $\zeta - A$ is even surjective for all $\zeta > 0$, hence $\rho(A) = (0, \infty)$.

PROOF: Consider the set

$$\Lambda := \{\zeta \in \mathbb{R} : 0 < \zeta < \infty \text{ and } \mathfrak{R}(\zeta - A) = H\}$$

and notice that $\Lambda \neq \emptyset$, since $\zeta_0 \in \Lambda$. Now, if $\zeta \in \Lambda$, then $\zeta \in \rho(A)$ by property (7.2), Since $\rho(A)$ is open, any neighborhood of ζ is a subset of $\rho(A)$. The intersection of this neighborhood with the real line is clearly in Λ and therefore Λ is open. On the other hand, let $\zeta_n \in \Lambda$ with $\zeta_n \to \zeta > 0$. For every $y \in H$ there exists an element $x_n \in \mathfrak{D}(A)$ such that

$$\zeta_n x_n - A x_n = y. \tag{7.3}$$

By using theorem 7.2, we deduce that $\|x_n\| \leq \zeta_n^{-1} \|y\| \leq c$ for some $c > 0$. We notice now that

$$\begin{aligned}\zeta_m \|x_n - x_m\| &\leq \|\zeta_m (x_n - x_m) - A(x_n - x_m)\| \\ &= |\zeta_n - \zeta_m| \, \|x_n\| \leq c \cdot |\zeta_n - \zeta_m|.\end{aligned}$$

Therefore $\{x_n\}$ is a Cauchy sequence. Let $x_n \to x$. Then by (7.3) $Ax_n \to \zeta x - y$. Since A is closed, $x \in \mathfrak{D}(A)$ and $\zeta x - Ax = y$. Therefore $\mathfrak{R}(\zeta - A) = H$ and $\zeta \in \Lambda$. Thus Λ is closed in $(0, \infty)$. Because Λ is non-empty as well as open and closed at the same time, we have $\Lambda = (0, \infty)$. □

Finally, for all $\zeta > 0$ and $y \in \mathfrak{R}(\zeta - A)$ the following estimation is valid:

$$\|(\zeta - A)^{-1} y\| \leq \zeta^{-1} \|y\| .$$

This is a consequence of the fact that $\zeta - A$ is closed and of the bounded-below theorem, which quotes that $\mathfrak{R}(\zeta - A)$ is closed. In fact, in this case the inverse operator

$$(\zeta - A)^{-1} : \mathfrak{R}(\zeta - A) \to \mathfrak{D}(A)$$

is bounded and has operator norm $\|(\zeta - A)^{-1}\| \leq \zeta^{-1}$, so the operator $(\zeta - A)^{-1}$ forms a contraction for all $\zeta \geq 1$.

Let us now switch to the main theorem of this section. By means of the notation of m-dissipativity, the *Lumer-Phillips theorem* classifies the linear operators being the infinitesimal generators of contractive C_0-semigroups.

THEOREM 7.5. Let A be a linear operator in a complex Hilbert space. Then A is m-disspipative if and only if A is the infinitesimal generator of a contractive C_0-semigroup.

PROOF: Let A be m-dissipative. Then theorem 7.3 quotes that A is densely defined and closed. By theorem 6.1, it is also to show that $(0,\infty)\subset\rho(A)$ and the resolvent estimate holds, that is, for every $\zeta>0$ we have $\|R(A,\zeta)\|\le\zeta^{-1}$. In fact, since $\mathfrak{R}(\zeta-A)=H$ for $\zeta>0$, inequality (7.2) implies that $(\zeta-A)^{-1}$ is a bounded linear operator, hence $\zeta\in\rho(A)$. Moreover, the resolvent estimate follows from (7.2). Thus the Hille-Yosida theorem yields that A is the infinitesimal generator of a C_0-semigroup of contractions.

Reversely, if A is the infinitesimal generator of a C_0-semigroup of contractions, then by theorem 6.1 we have $(0,\infty)\subset\rho(A)$ and therefore $\mathfrak{R}(\zeta-A)=H$ for all $\zeta>0$. In addition, if $x\in\mathfrak{D}(A)$, then

$$|\langle S(t)x,x\rangle| \;\le\; \|S(t)x\|\,\|x\| \;\le\; \|x\|^2$$

and

$$\operatorname{Re}\langle S(t)x-x,x\rangle \;=\; \operatorname{Re}\langle S(t)x,x\rangle-\|x\|^2 \;\le\; 0.$$

Dividing this line by $t>0$ and letting $t\downarrow 0$ yields $\operatorname{Re}\langle Ax,x\rangle\le 0$, which shows that A is dissipative. In summary, the operator A proves to be m-dissipative. □

Recall that to each densely defined linear operator A acting in a Hilbert H there is a linear operator A^*, called the *adjoint operator* of A, such that $\langle Ax,y\rangle=\langle x,A^*y\rangle$ for all $x\in\mathfrak{D}(A)$ and $y\in\mathfrak{D}(A^*)$.

Having the concept of Hilbert adjoint available, we obtain a further criterion for A to be the generator of a contraction C_0-semigroup:

THEOREM 7.6. Let A be a densely defined linear operator in a complex Hilbert space H. If both, A and A^* are dissipative, then the operator A is the generator of a C_0-semigroup of contractions.

PROOF. From theorem 7.5 it is sufficient to show that $\mathfrak{R}(1-A)=X$. Since A is dissipative and closed, $\mathfrak{R}(1-A)$ is a closed subspace of X. If $\mathfrak{R}(1-A)\neq X$ then there exists $x\in H$, $x\neq 0$ such that $\langle x,x-Ax\rangle=0$ for $x\in\mathfrak{D}(A)$. This implies $x-Ax=0$. Since A^* is also dissipative, it follows from theorem 7.2 that $x=0$, contradicting the construction of x. □

We have formulated the Lumer-Phillips theorem in the setting of a complex Hilbert space. Let us finally mention that this theorem is also valid when A is a linear operator acting in a complex Banach space. In order to quote an analogous theorem, we have to substitute the inner product in a Hilbert space by a similar mapping, where *dual pairs* are used.

Let X be a Banach space and X^* its dual space, that is, the linear space of continuous linear functionals $x^*: X \to \mathbb{C}$. We denote the value of $x^* \in X^*$ at $x \in X$ by either $\langle x^*, x\rangle$ or $\langle x, x^*\rangle$. For any $x \in X$ let $F(x) \subset X^*$ be the *duality set* defined by

$$F(x) = \{x^* \in X^* : \langle x^*, x\rangle = \|x\|^2 = \|x^*\|^2\}.$$

It follows from the *Hahn-Banach theorem* that $F(x) \neq \emptyset$ for $x \in X$. Then a linear operator A acting in X is called *dissipative* if for every element $x \in \mathfrak{D}(A)$ there exists $x^* \in F(x)$ such that $\operatorname{Re}\langle x^*, x\rangle \leq 0$.

If $X = H$ is a Hilbert space, however, the set $F(x)$ consists of the singleton $\{x\}$ after identifying the element $x \in H$ with the continuous linear functional $f_x := \langle x, \cdot\rangle \in H^*$.

8. Applications to Partial Differential Operators

The previously presented theory has many applications, especially with respect to linear partial differential operators.

1. We consider the Laplacian Δ in the entire Euclidean space $\mathbb{R}^d$. Then theorem 7.5 ensures that the realization A_Δ in Hilbert space $X = L^2(\mathbb{R}^d)$ of square-integrable functions $u: \mathbb{R}^d \to \mathbb{C}$ and with domain of definition

$$\mathfrak{D}(A_\Delta) := H^2(\mathbb{R}^d)$$

and action $Au := \Delta u$ for $u \in \mathfrak{D}(A_\Delta)$ is the infinitesimal generator of a strongly continuous semigroup $\{G(t)\}_{t\geq 0}$, denoted the *Gauss-Weierstrass semigroup.* In fact, it is not hard to show that the operator A_Δ is dissipative, and m-dissipativity follows from self-adjointness of A_Δ.

It is a consequence of theorem 7.5 that there is a uniquely defined solution $u \in C^1(I \times \mathbb{R}^d, \mathbb{R})$ for the homogeneous Cauchy problem

$$\begin{cases} \dot{u} = A_\Delta u, \\ u(x,0) = u_0(x) \text{ for } x \in \mathbb{R}^d \end{cases}$$

with initial mapping $u_0 \in H^2(\mathbb{R}^d)$. Furthermore, it is possible to represent the semigroup $\{G(t)\}_{t\geq 0}$ explicitly by means of the *heat kernel* function

$$\Phi_d(t,x) := \frac{1}{(4\pi t)^{d/2}}\, e^{-|x|^2/4t},$$

that is, for all $t > 0$ and $x \in \mathbb{R}^d$ we have

$$(G(t)\, u_0)\,(x) \;=\; \frac{1}{(4\pi t)^{d/2}} \int_{\mathbb{R}^d} e^{-\frac{|x-y|^2}{4t}}\, u_0(y)\, dy \;=\; (\Phi_d(t) * u_0)\,(x).$$

2. Next we consider the Laplacian Δ in a bounded domain $G \subset \mathbb{R}^d$. The differential expression gives rise to a linear operator A_Δ in Hilbert space $L^2(G)$ and with domain of definition

$$\mathfrak{D}(A_\Delta) \;:=\; \{u \in L^2(G) : u \in H_0^1(G),\, \Delta u \in L^2(G)\},$$

that is, homogeneous Dirichlet boundary conditions are incorporated into the domain of definition of A_Δ. If the boundary of G is sufficiently smooth, then it follows from elliptic regularity theory that actually $\mathfrak{D}(A_\Delta) = H_0^1(G) \cap H^2(G)$ holds.

By means of GREEN's formula it can be shown that the operator A_Δ is dissipative. Moreover, using Hilbert space methods and especially the Lax-Milgram lemma, A_Δ fulfills the range condition and hence proves to be m-dissipative. Thus it is the infinitesimal generator of a C_0-semigroup.

In conclusion, the homogeneous Cauchy problem

$$\begin{cases} \dot{u} \;=\; A_\Delta u, \\ u(x,t) \;=\; 0 \text{ for } t > 0 \text{ and } x \in \partial G, \\ u(x,0) \;=\; u_0(x) \text{ for } x \in \overline{G} \end{cases}$$

has a unique solution $u \in C^1(I \times \overline{G}, \mathbb{R})$. But for general domains it is not possible to represent the semigroup by a kernel function.

3. Let $X = L^2(\mathbb{R})$ and $Au := i\,u''$ for $u \in \mathfrak{D}(A) := H^2(\mathbb{R})$. Since the operator $u \in H^2(\mathbb{R}) \mapsto u''$ is self-adjoint, it follows that the operator A and its adjoint A^* fulfill the properties $\langle Au, u\rangle \in i\mathbb{R}$ and $\langle A^*u, u\rangle \in i\mathbb{R}$, hence they are dissipative. Thus the operator A is the generator of a contraction semigroup by theorem 7.6.

4. Let $X = L^2(0,1)$ and $Au := u'$ for u in the domain of definition

$$\mathfrak{D}(A) \;:=\; \{u \in H^1(0,1) : u(1) = 0\}.$$

Then A is m-dissipative and hence the generator of a C_0-semigroup of contractions. Indeed, we have

$$\langle Au, u\rangle \;=\; \int_0^1 u' u\, dx \;=\; -\frac{1}{2}u^2(0) \;\le\; 0.$$

To verify the range condition, consider the differential equation

$u - u' = f$ for $f \in L^2(0,1)$ with solution

$$u(t) = e^t - \int_0^t f(s)\, ds, \quad (t \in [0,1]).$$

5. Let $X = C_{ub}(\mathbb{R})$, the Banach space of bounded and uniformly continuous functions $u : \mathbb{R} \to \mathbb{R}$ and $Au := u'$ for u in the domain of definition

$$\mathfrak{D}(A) := \{u \in C_{ub}(\mathbb{R}) : u \in C^1(\mathbb{R}) \text{ and } u' \in C_{ub}(\mathbb{R})\}.$$

Then A is the generator of the *translation* C_0-semigroup $\{T(t)\}_{t \geq 0}$ in $\mathbb{R}$ given by $T(t)u(s) := u(s+t)$ for $u \in C_{ub}(\mathbb{R})$ and $t, s \in \mathbb{R}$.

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